

RESEARCH PAPER 3

A Research Study of Ambulance Operations and Best Practice Considerations for Emergency Medical Services Personnel

- Department of Homeland Security (DHS)

Abstract:

The emergency medical services (EMS) community faces many challenges in providing patient care while maintaining the safety of their patients and themselves. One factor that influences patient care and safety is the ability of the EMS provider operating the ambulance to quickly but safely maneuver to the site of the medical emergency and subsequently transport the patient(s) to the hospital. The Department of Homeland Security (DHS) and the Emergency Medical Services Community identified a need to research best practices for ambulance operators and identify safety gaps. This research report coincides with the DHS Science and Technology Directorate's First Responders Group and the Resilient Systems Division's partnership with the National Institute of Standards and Technology, the National Institute for Occupational Safety and Health, BMT Designers and Planners (D&P), and Carlow International's project to develop ambulance safety and design standards and recommendations. The project provides design guidance for ambulance patient compartments for crashworthiness, patient safety and comfort, and EMS provider safety and performance. This research report summarizes the efforts of this team to identify best practices and considerations for use and consideration by the EMS community. A team comprised of D&P and Carlow International used human performance requirements analysis, literature reviews and driver (ambulance operator) interviews to identify ambulance operator tasks required in responding to an incident, transporting patients and potential inhibitors to performance and safety. This research report summarizes best practice information received and consolidates it in an appendix for consideration by the EMS community.